

Project Objective

The objective of this investigation was to act as a SOC analyst at a financial services company called FinanceCore Ltd and investigate a SIEM alert flagging unusual weekend activity on host FINCORE-WS04. The goal was to use Splunk as a real SIEM tool to search, filter and correlate log data, identify all attack phases, map findings to MITRE ATT&CK techniques, produce a professional investigation report and document personal lessons learned throughout the process. This investigation forms the capstone piece of my self-directed Phase 4 SOC analyst skills portfolio.

Raw Analysis

The FINCIRE-WS04 host was online at around 12:00 AM. The host attempted to reach a Microsoft domain, as shown by six DNS queries. This was followed by five failed login attempts to the mthembu user account from IP address 41.185.28.93.

A successful login to the mthembu account then occurred. The account was subsequently used to access several sensitive files, including client accounts, client transactions, and internal staff salaries. A file archive was then created using the mthembu account.

The antivirus software was disabled by the mthembu account, indicating possible defense evasion. The host then established an outbound connection to 196.203.44.81 over port 443 (HTTPS). A new user account, svc_update01, was created by the mthembu account and assigned administrator privileges. A scheduled task was also created using the mthembu account, which could indicate persistence.

There was then a successful login to the svc_update01 account from IP address 196.203.44.81. Another outbound connection was established to the same IP address over port 4444. The registry was subsequently modified, followed by another successful login to the svc_update01 account. Several files were then accessed, and the activity ended with another outbound connection to 196.203.44.81 over port 4444. Suspicious/abnormal activity

The incident contains several indicators of suspicious or abnormal activity.

First, there were multiple failed login attempts against the mthembu account. The attempts occurred approximately four seconds apart, which could indicate an automated password-

guessing or brute-force attack rather than manual attempts by a human. The subsequent successful login increases the concern that the account credentials may have been compromised. After the successful login, the account was used to access sensitive financial and employee information. This is suspicious because the activity involved files containing confidential data.

An archive was also created after the sensitive files were accessed. Creating an archive can be legitimate, but in this context, it is suspicious because compressing multiple files into a single archive can make them easier to transfer or potentially exfiltrate.

The antivirus software was then disabled by the mthembu account. This is an indicator of defense evasion, as an attacker may disable security controls to reduce the likelihood of detection and allow malicious activity to continue. A new account, svc_update01, was created and assigned administrator privileges. This is suspicious because administrative privileges provide extensive access to the system and its resources. If the account was created by an attacker, it could have been used to access additional sensitive information or make further system changes. The creation of a scheduled task is also suspicious and could indicate persistence. Scheduled tasks can allow malicious programs or commands to execute automatically at a later time, potentially allowing an attacker to maintain access even after their initial session ends.

Finally, repeated outbound connections to 196.203.44.81 over port 4444 are suspicious, particularly because the same external IP address was associated with a successful login to the newly created administrator account. Port 4444 is commonly associated with certain remote-access and penetration-testing tools, although the port number alone does not prove that malicious software was being used. The repeated connections should therefore be investigated further.

What do I think the attacker's ultimate end goal was at FinanceCore Ltd?

Based on the available events, the attacker's ultimate goal cannot be confirmed with certainty because there is no clear evidence of completed data exfiltration. However, the activity suggests an attempt to compromise the confidentiality and integrity of FinanceCore Ltd's systems and sensitive information.

Therefore, the most likely objective was to gain and maintain unauthorized privileged access while compromising the confidentiality and integrity of FinanceCore Ltd's data and systems.

However, further evidence would be required to determine whether the ultimate goal was data theft, system manipulation, financial fraud, or another objective.

SPLUNK EVIDENCE

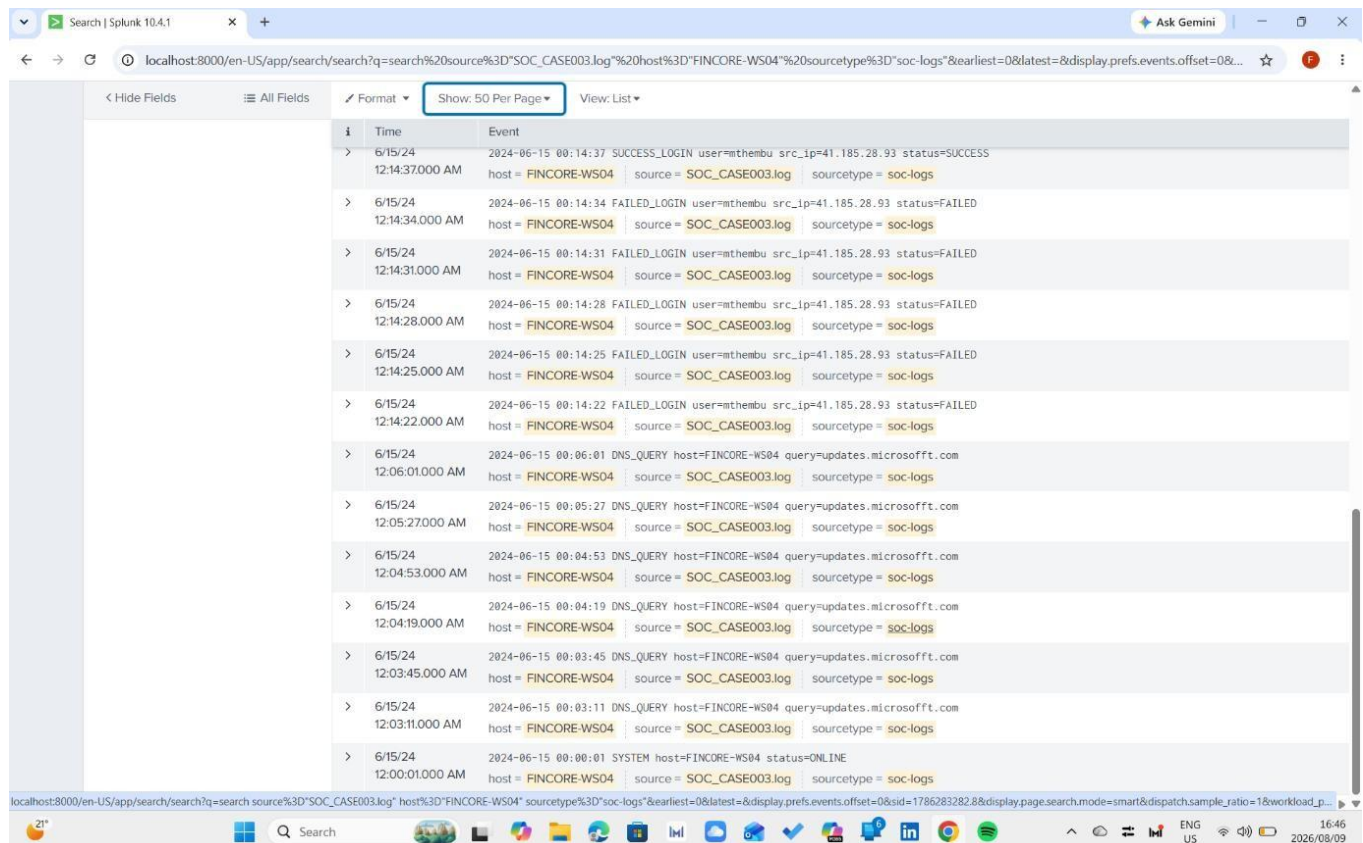
Attachment 1

Full Event Timeline

Objective: Load all Case 003 log events into Splunk and view the complete attack timeline in chronological order to get an initial picture of what occurred on FINCOREWS04 before running any specific queries.

Query used: `source="SOC_CASE003.log" | table`

`_time, _raw`



#	Time	Event
>	6/15/24 12:14:37.000 AM	2024-06-15 00:14:37 SUCCESS_LOGIN user=mthembu src_ip=41.185.28.93 status=SUCCESS host = FINCORE-WS04 source = SOC_CASE003.log sourcetype = soc-logs
>	6/15/24 12:14:34.000 AM	2024-06-15 00:14:34 FAILED_LOGIN user=mthembu src_ip=41.185.28.93 status=FAILED host = FINCORE-WS04 source = SOC_CASE003.log sourcetype = soc-logs
>	6/15/24 12:14:31.000 AM	2024-06-15 00:14:31 FAILED_LOGIN user=mthembu src_ip=41.185.28.93 status=FAILED host = FINCORE-WS04 source = SOC_CASE003.log sourcetype = soc-logs
>	6/15/24 12:14:28.000 AM	2024-06-15 00:14:28 FAILED_LOGIN user=mthembu src_ip=41.185.28.93 status=FAILED host = FINCORE-WS04 source = SOC_CASE003.log sourcetype = soc-logs
>	6/15/24 12:14:25.000 AM	2024-06-15 00:14:25 FAILED_LOGIN user=mthembu src_ip=41.185.28.93 status=FAILED host = FINCORE-WS04 source = SOC_CASE003.log sourcetype = soc-logs
>	6/15/24 12:14:22.000 AM	2024-06-15 00:14:22 FAILED_LOGIN user=mthembu src_ip=41.185.28.93 status=FAILED host = FINCORE-WS04 source = SOC_CASE003.log sourcetype = soc-logs
>	6/15/24 12:06:01.000 AM	2024-06-15 00:06:01 DNS_QUERY host=FINCORE-WS04 query=updates.microsoft.com host = FINCORE-WS04 source = SOC_CASE003.log sourcetype = soc-logs
>	6/15/24 12:05:27.000 AM	2024-06-15 00:05:27 DNS_QUERY host=FINCORE-WS04 query=updates.microsoft.com host = FINCORE-WS04 source = SOC_CASE003.log sourcetype = soc-logs
>	6/15/24 12:04:53.000 AM	2024-06-15 00:04:53 DNS_QUERY host=FINCORE-WS04 query=updates.microsoft.com host = FINCORE-WS04 source = SOC_CASE003.log sourcetype = soc-logs
>	6/15/24 12:04:19.000 AM	2024-06-15 00:04:19 DNS_QUERY host=FINCORE-WS04 query=updates.microsoft.com host = FINCORE-WS04 source = SOC_CASE003.log sourcetype = soc-logs
>	6/15/24 12:03:45.000 AM	2024-06-15 00:03:45 DNS_QUERY host=FINCORE-WS04 query=updates.microsoft.com host = FINCORE-WS04 source = SOC_CASE003.log sourcetype = soc-logs
>	6/15/24 12:03:11.000 AM	2024-06-15 00:03:11 DNS_QUERY host=FINCORE-WS04 query=updates.microsoft.com host = FINCORE-WS04 source = SOC_CASE003.log sourcetype = soc-logs
>	6/15/24 12:00:01.000 AM	2024-06-15 00:00:01 SYSTEM host=FINCORE-WS04 status=ONLINE host = FINCORE-WS04 source = SOC_CASE003.log sourcetype = soc-logs

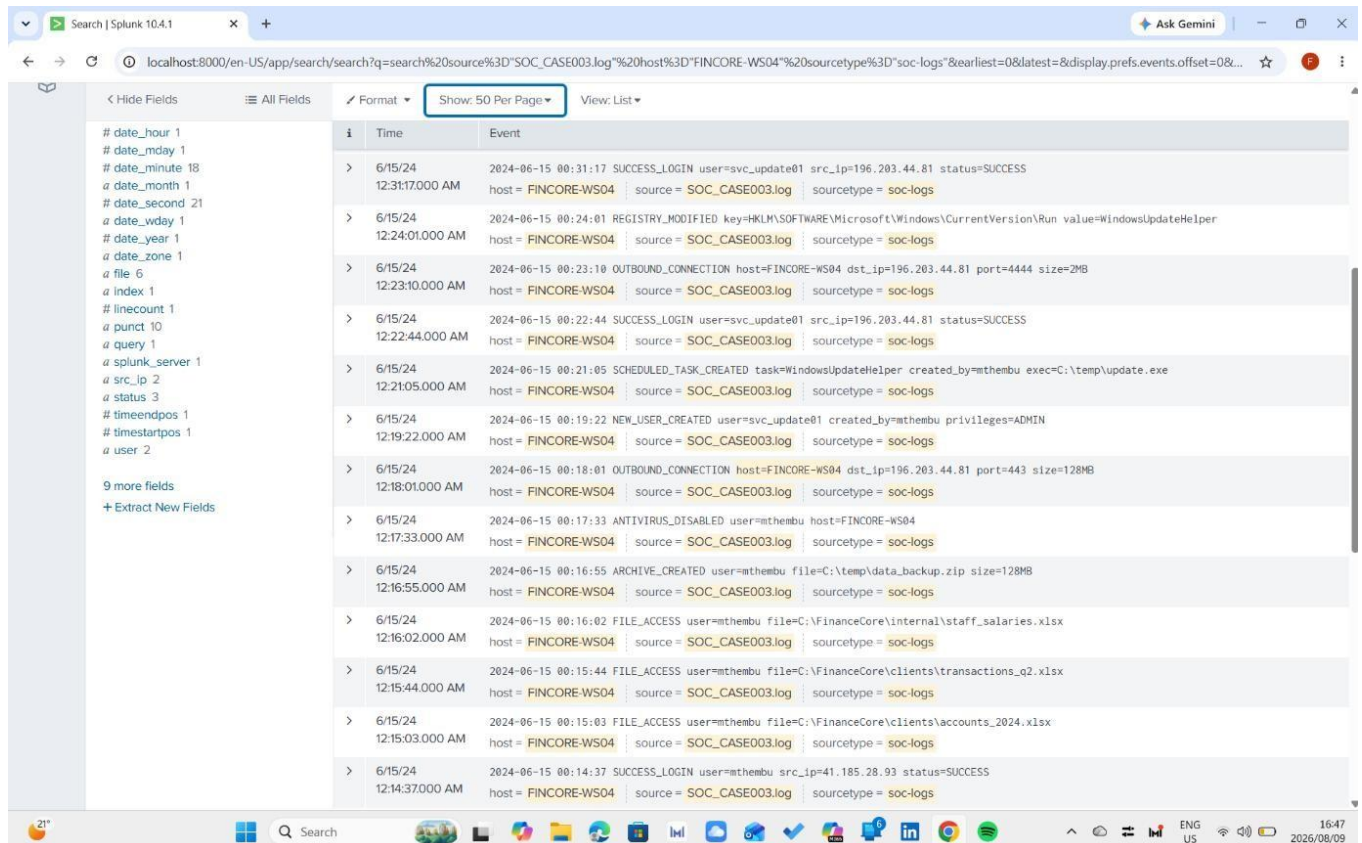
Results: 28 events were returned covering the period from 00:00:01 to 00:33:55 on 15 June 2024. The timeline revealed a system heartbeat at midnight, followed by repeated DNS queries to updates.microsoft.com, then failed and successful logins, file access, archive creation, antivirus disabling, outbound connections, new user creation, scheduled task creation, registry modification and further file access and outbound connections. The host field correctly showed FINCORE-WS04 confirming proper data ingestion.

What this proved: The full attack chain was visible in one view, from pre-attack beaconing all the way to active reverse shell communication at the end of the timeline.

Attachment 2

Full Event Timeline Continued

Objective: Continue reviewing the second half of the timeline focusing on postcompromise activity including persistence mechanisms and command and control traffic.



#	Time	Event
>	6/15/24 12:31:17.000 AM	2024-06-15 00:31:17 SUCCESS_LOGIN user=svc_update01 src_ip=196.203.44.81 status=SUCCESS host = FINCORE-WS04 source = SOC_CASE003.log sourcetype = soc-logs
>	6/15/24 12:24:01.000 AM	2024-06-15 00:24:01 REGISTRY_MODIFIED key=HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Run value=WindowsUpdateHelper host = FINCORE-WS04 source = SOC_CASE003.log sourcetype = soc-logs
>	6/15/24 12:23:10.000 AM	2024-06-15 00:23:10 OUTBOUND_CONNECTION host=FINCORE-WS04 dst_ip=196.203.44.81 port=4444 size=2MB host = FINCORE-WS04 source = SOC_CASE003.log sourcetype = soc-logs
>	6/15/24 12:22:44.000 AM	2024-06-15 00:22:44 SUCCESS_LOGIN user=svc_update01 src_ip=196.203.44.81 status=SUCCESS host = FINCORE-WS04 source = SOC_CASE003.log sourcetype = soc-logs
>	6/15/24 12:21:05.000 AM	2024-06-15 00:21:05 SCHEDULED_TASK_CREATED task=WindowsUpdateHelper created_by=mthembu exec=C:\temp\update.exe host = FINCORE-WS04 source = SOC_CASE003.log sourcetype = soc-logs
>	6/15/24 12:19:22.000 AM	2024-06-15 00:19:22 NEW_USER_CREATED user=svc_update01 created_by=mthembu privileges=ADMIN host = FINCORE-WS04 source = SOC_CASE003.log sourcetype = soc-logs
>	6/15/24 12:18:01.000 AM	2024-06-15 00:18:01 OUTBOUND_CONNECTION host=FINCORE-WS04 dst_ip=196.203.44.81 port=443 size=128MB host = FINCORE-WS04 source = SOC_CASE003.log sourcetype = soc-logs
>	6/15/24 12:17:33.000 AM	2024-06-15 00:17:33 ANTIVIRUS_DISABLED user=mthembu host=FINCORE-WS04 host = FINCORE-WS04 source = SOC_CASE003.log sourcetype = soc-logs
>	6/15/24 12:16:55.000 AM	2024-06-15 00:16:55 ARCHIVE_CREATED user=mthembu file=C:\temp\data_backup.zip size=128MB host = FINCORE-WS04 source = SOC_CASE003.log sourcetype = soc-logs
>	6/15/24 12:16:02.000 AM	2024-06-15 00:16:02 FILE_ACCESS user=mthembu file=C:\FinanceCore\internal\staff_salaries.xlsx host = FINCORE-WS04 source = SOC_CASE003.log sourcetype = soc-logs
>	6/15/24 12:15:44.000 AM	2024-06-15 00:15:44 FILE_ACCESS user=mthembu file=C:\FinanceCore\clients\transactions_q2.xlsx host = FINCORE-WS04 source = SOC_CASE003.log sourcetype = soc-logs
>	6/15/24 12:15:03.000 AM	2024-06-15 00:15:03 FILE_ACCESS user=mthembu file=C:\FinanceCore\clients\accounts_2024.xlsx host = FINCORE-WS04 source = SOC_CASE003.log sourcetype = soc-logs
>	6/15/24 12:14:37.000 AM	2024-06-15 00:14:37 SUCCESS_LOGIN user=mthembu src_ip=41.185.28.93 status=SUCCESS host = FINCORE-WS04 source = SOC_CASE003.log sourcetype = soc-logs

Results: The second half of the timeline confirmed scheduled task creation, registry modification, backdoor account logins and final outbound connections on port 4444. Seeing the attacker login using the newly created svc_update01 account from the same external IP confirmed that the backdoor account was immediately used for continued access.

What this proved: The attacker did not leave after initial access. They embedded themselves deeply into FINCORE-WS04 using three separate persistence mechanisms to ensure continued access even if the original mthembu account was disabled.

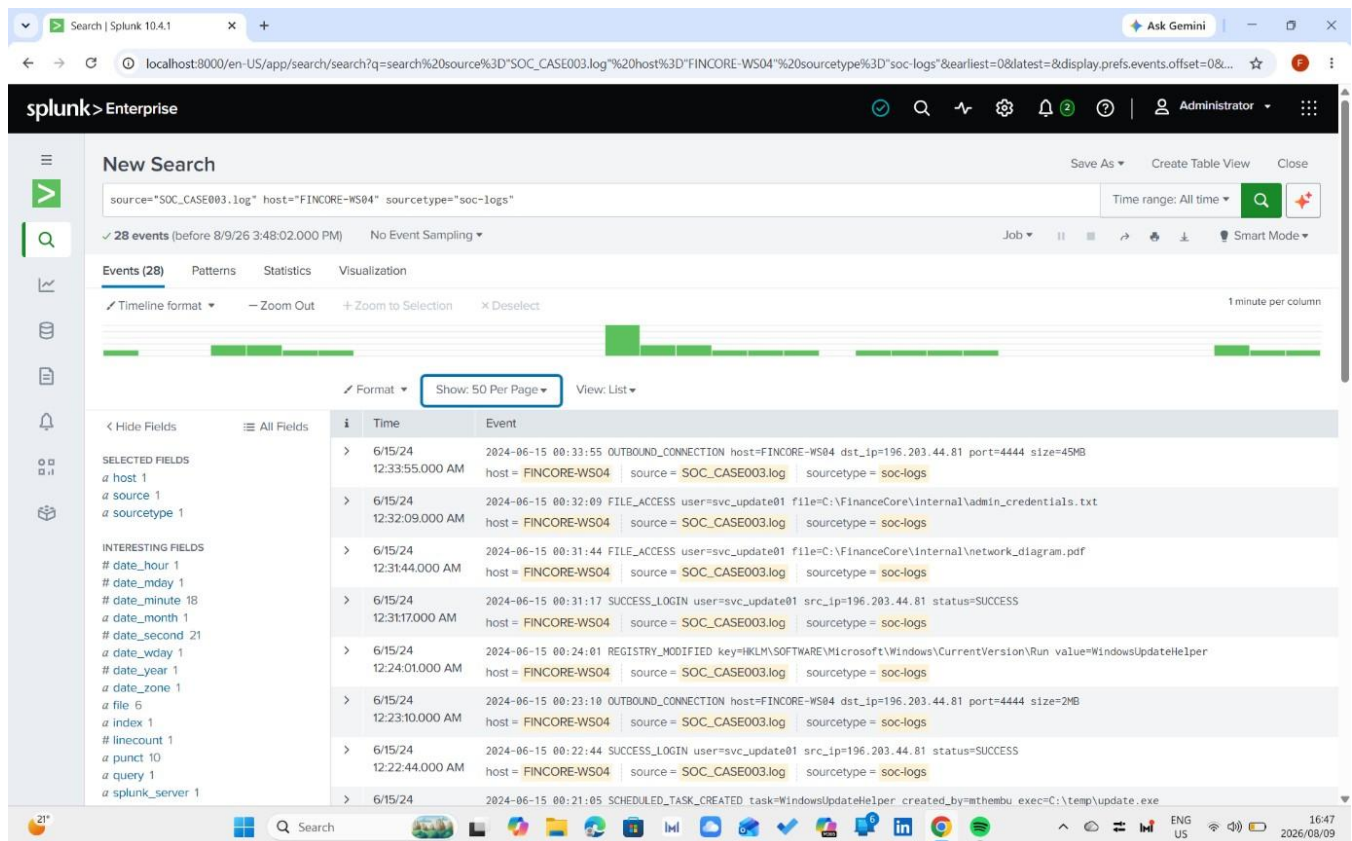
Attachment 3

Full Timeline with Final Outbound Connection

Objective: View the complete 28-event timeline in Splunk with the full attack visible from start to finish, specifically confirming the final outbound connection that concluded the active attack phase on FINCORE-WS04.

Query used: `source="SOC_CASE003.log" host="FINCORE-WS04"`

`sourcetype="soc-logs"`



Results: 28 events confirmed with host correctly showing FINCORE-WS04. The final event at 00:33:55 showed:

OUTBOUND_CONNECTION host=FINCORE-WS04 dst_ip=196.203.44.81 port=4444 size=45MB

This was the last recorded attacker action - a 45MB data transfer via the reverse shell port 4444 to the same C2 server used throughout the entire attack.

What this proved: The attack did not end after the initial 128MB exfiltration via port 443. The attacker returned via the backdoor svc_update01 account, accessed additional sensitive files including admin_credentials.txt and network_diagram.pdf, and exfiltrated a further 45MB through the established reverse shell. The network diagram and admin credentials accessed in this final phase suggest the attacker was conducting internal reconnaissance, potentially planning lateral movement to other machines on the FinanceCore network beyond FINCORE-WS04.

Attachment 4

Event Type Breakdown

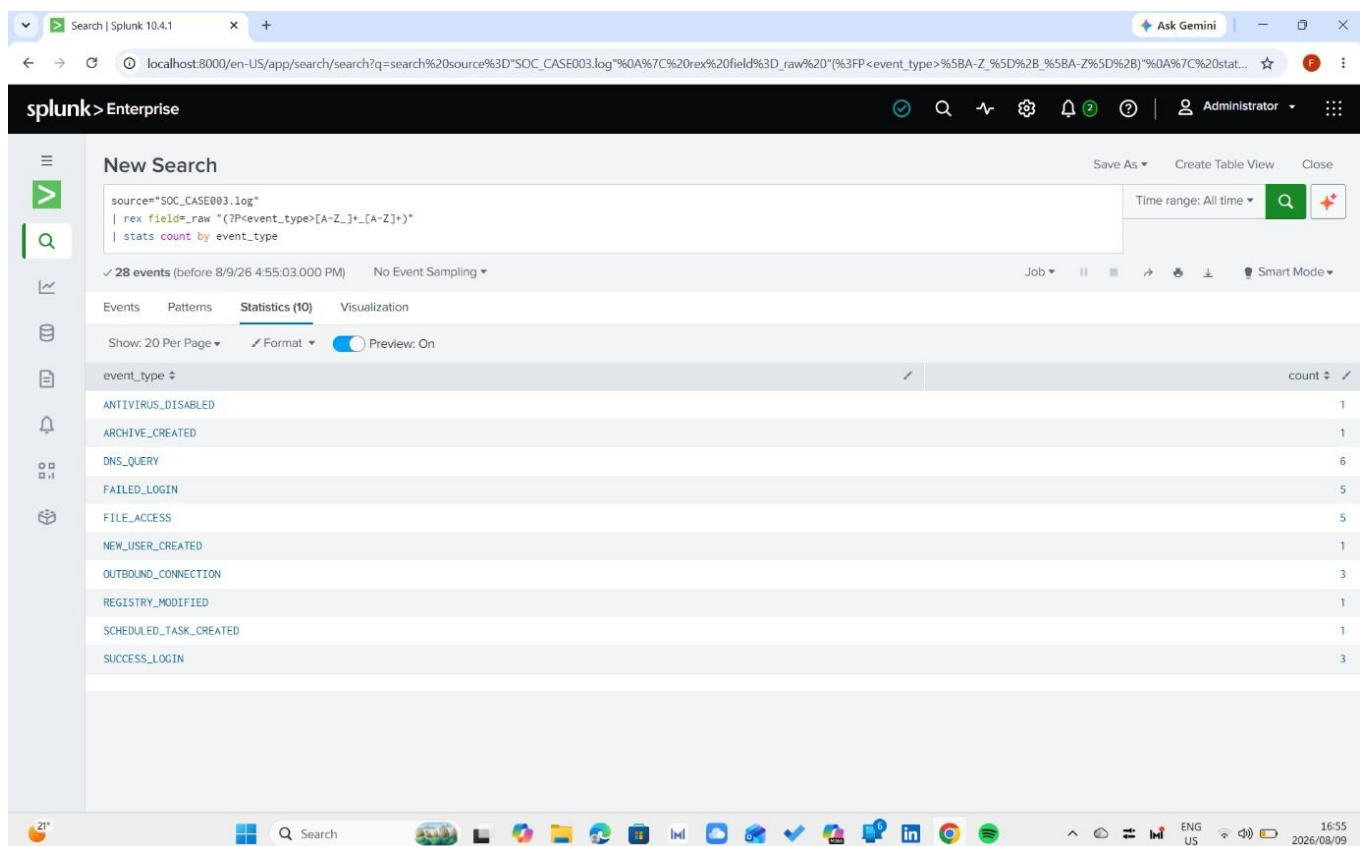
Objective: Use SPL to extract and count all event types across the 28 events to understand the full scope and variety of attack techniques used in one view.

Query used:

```
source="SOC_CASE003.log"
```

```
| rex field=_raw "(?P<event_type>[A-Z]+_[A-Z]+)"
```

```
| stats count by event_type
```



Results:

ANTIVIRUS_DISABLED: 1

ARCHIVE_CREATED: 1

DNS_QUERY: 6

FAILED_LOGIN: 5

FILE_ACCESS: 5

NEW_USER_CREATED: 1

OUTBOUND_CONNECTION: 3

REGISTRY_MODIFIED: 1

SCHEDULED_TASK_CREATED: 1

SUCCESS_LOGIN: 3

What this proved: This was the most complex incident I investigated across all three cases. Ten different event types covering six distinct attack phases confirms this was an advanced multi-stage attack, not opportunistic. The variety of event types shows the attacker had clear objectives and executed them methodically.

Attachment 5

DNS Beaconsing Evidence

Objective: Isolate all DNS query events to prove the pre-attack beaconsing pattern and confirm the typosquatted domain.

Query used:

source="SOC_CASE003.log" DNS_QUERY

| table _time, _raw

The screenshot displays the Splunk Enterprise web interface. The search bar contains the query: `source="SOC_CASE003.log" DNS_QUERY | table _time, _raw`. The search results show 6 events. The table below represents the data shown in the search results.

_time	_raw
2024-06-15 00:05:27	2024-06-15 00:05:27 DNS_QUERY host=FINCORE-WS04 query=updates.microsoft.com
2024-06-15 00:04:53	2024-06-15 00:04:53 DNS_QUERY host=FINCORE-WS04 query=updates.microsoft.com
2024-06-15 00:04:19	2024-06-15 00:04:19 DNS_QUERY host=FINCORE-WS04 query=updates.microsoft.com
2024-06-15 00:03:45	2024-06-15 00:03:45 DNS_QUERY host=FINCORE-WS04 query=updates.microsoft.com
2024-06-15 00:03:11	2024-06-15 00:03:11 DNS_QUERY host=FINCORE-WS04 query=updates.microsoft.com
2024-06-15 00:06:01	2024-06-15 00:06:01 DNS_QUERY host=FINCORE-WS04 query=updates.microsoft.com

Results: Six DNS queries all targeting updates.microsoft.com between 00:03:11 and 00:06:01. Timestamps showed exact 34-second intervals between each query 00:03:11, 00:03:45, 00:04:19, 00:04:53, 00:05:27, 00:06:01.

What this proved: Machine-precision 34-second intervals confirm automated malware beaconing. The domain updates.microsoft.com contains a deliberate double f, typosquatting designed to impersonate the legitimate Microsoft update domain. This confirmed that FINCORE-WS04 was already infected with malware before the brute force attack at 00:14 meaning the initial compromise happened before the visible attack window began.

Attachment 6

Authentication Status and Exfiltration Evidence

Objective: Two queries run simultaneously, first to break down login statuses to prove the brute force pattern, then to correlate the archive creation with outbound connection sizes to confirm data exfiltration.

Query 1:

```
source="SOC_CASE003.log"
```

```
| rex field=_raw "status=(?P\w+)"
```

```
| stats count by status
```


The screenshot shows the Splunk Enterprise web interface. The search bar contains the following query:

```
source="SOC_CASE003.log"
| rex field=_raw "status=(?P<status>\w+)"
| stats count by status
```

The search results are displayed in a table with the following data:

status	count
FAILED	5
ONLINE	1
SUCCESS	3

The interface also shows a sidebar with navigation icons, a top navigation bar with the Splunk logo and user information, and a bottom taskbar with various application icons.

Results: FAILED: 5, SUCCESS: 3, ONLINE: 1

What this proved: Five failed attempts before success confirmed the brute force attack. The three successes represented mthembu's compromised login and two subsequent logins from svc_update01, all from the same external IP 196.203.44.81

Attachment 7

Query 2:

```
source="SOC_CASE003.log" ARCHIVE_CREATED OR OUTBOUND_CONNECTION
```

```
| table _time, _raw
```

New Search

source="SOC_CASE003.log" ARCHIVE_CREATED OR OUTBOUND_CONNECTION
| table _time, _raw

4 events (before 8/9/26 5:01:13.000 PM) No Event Sampling

Events Patterns **Statistics (4)** Visualization

Show: 20 Per Page Format Preview: On

_time	_raw
2024-06-15 00:23:10	2024-06-15 00:23:10 OUTBOUND_CONNECTION host=FINCORE-WS04 dst_ip=196.203.44.81 port=4444 size=2MB
2024-06-15 00:18:01	2024-06-15 00:18:01 OUTBOUND_CONNECTION host=FINCORE-WS04 dst_ip=196.203.44.81 port=443 size=128MB
2024-06-15 00:16:55	2024-06-15 00:16:55 ARCHIVE_CREATED user=mthembu file=C:\temp\data_backup.zip size=128MB
2024-06-15 00:33:55	2024-06-15 00:33:55 OUTBOUND_CONNECTION host=FINCORE-WS04 dst_ip=196.203.44.81 port=4444 size=45MB

Results:

00:16:55 ARCHIVE_CREATED size=128MB

00:18:01 OUTBOUND_CONNECTION port=443 size=128MB 00:23:10

OUTBOUND_CONNECTION port=4444 size=2MB

00:33:55 OUTBOUND_CONNECTION port=4444 size=45MB

What this proved: The 128MB archive created at 00:16:55 was transferred outbound exactly 66 seconds later at 00:18:01 identical file sizes confirming the archive was the data that was exfiltrated. Total data exfiltrated was 175MB of sensitive FinanceCore financial data including client accounts, transactions and staff salaries.

Attachment 8

Persistence Mechanisms Evidence

Objective: Isolate all three persistence mechanisms in a single query to prove the attacker embedded multiple backdoors into FINCORE-WS04 to maintain long-term access even after detection.

Query used:

source="SOC_CASE003.log" SCHEDULED_TASK_CREATED OR REGISTRY_MODIFIED

OR NEW_USER_CREATED

| table _time, _raw

The screenshot shows the Splunk Enterprise web interface. The search bar contains the query: `source="SOC_CASE003.log" SCHEDULED_TASK_CREATED OR REGISTRY_MODIFIED OR NEW_USER_CREATED | table _time, _raw`. The search results show 3 events. The first event is a REGISTRY_MODIFIED event at 00:24:01. The second event is a SCHEDULED_TASK_CREATED event at 00:21:05. The third event is a NEW_USER_CREATED event at 00:19:22.

_time	_raw
2024-06-15 00:24:01	2024-06-15 00:24:01 REGISTRY_MODIFIED key=HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Run value=WindowsUpdateHelper
2024-06-15 00:21:05	2024-06-15 00:21:05 SCHEDULED_TASK_CREATED task=WindowsUpdateHelper created_by=mthembu exec=C:\temp\update.exe
2024-06-15 00:19:22	2024-06-15 00:19:22 NEW_USER_CREATED user=svc_update01 created_by=mthembu privileges=ADMIN

Results:

00:19:22 NEW_USER_CREATED user=svc_update01 created_by=mthembu privileges=ADMIN

00:21:05 SCHEDULED_TASK_CREATED task=WindowsUpdateHelper created_by=mthembu
exec=C:\temp\update.exe

00:24:01 REGISTRY_MODIFIED
key=HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Run
value=WindowsUpdateHelper

What this proved: Three separate persistence mechanisms were confirmed in one query. The backdoor admin account `svc_update01` gives the attacker a login that survives password resets on the `mthembu` account. The scheduled task `WindowsUpdateHelper` executes malware automatically at set intervals. The registry `Run` key ensures the malware launches every time Windows starts even after a reboot. Together these three mechanisms mean the attacker remains present on `FINCOREWS04` regardless of which individual backdoor defenders discover first, removing one leaves two others active. This is the hallmark of an experienced and deliberate attacker who anticipated discovery.

What do I think the attacker's ultimate end goal was at FinanceCore Ltd? Updated

"Based on the available events, the attacker's ultimate goal was to compromise the confidentiality of FinanceCore Ltd's sensitive financial data. Evidence from Splunk confirmed that 175MB of data was successfully exfiltrated — a 128MB archive transferred via port 443 followed by a further 45MB via port 4444. Additionally, the attacker accessed `admin_credentials.txt` and `network_diagram.pdf` suggesting the ultimate goal extended beyond FINCORE-WS04 to potential lateral movement across the entire FinanceCore network."

Challenges I Faced

The biggest challenge in this investigation was the Splunk disk space issue that occurred during setup. My C: drive had only 1.99GB of free space and Splunk requires a minimum of 5000MB to run searches. I had to troubleshoot this by adjusting Splunk's minimum free space threshold in the General Settings, which taught me that real SOC environments require constant disk space management and that Splunk administrators set up alerts when storage drops below threshold.

A second challenge was the host field issue where Splunk automatically tagged all events with my laptop name `Fortunate-PC` instead of the actual victim host `FINCOREWS04`. I solved this by manually changing the host field value during the data upload process. In a real SOC environment different machines send logs to a central Splunk server so this would not occur, but it was a useful lesson in understanding how Splunk handles host metadata.

A third challenge was saving log files correctly. When saving in Notepad the default file type is `.txt` and Splunk stores the source name including the extension. This caused search queries to fail when I used `.log` in the source field but the file was saved as `.txt`. I learned to always change the Save As Type to All Files and specify the full filename including the correct extension.

What I Initially Missed and What It Taught Me

The most significant analytical gap I had in this investigation was missing the pre-attack beaconing phase initially. I identified the brute force attack, defense evasion, persistence and exfiltration correctly, but I did not immediately recognise the DNS queries at the top of the logs as a pre-attack C2 indicator. I initially thought of beaconing in the context of network traffic patterns I had studied but did not connect it to the DNS query pattern in this log format.

What this taught me is that real attackers rarely announce their presence. The most dangerous phase of an attack is when malware establishes C2 before any visible intrusion

begins to appear as something as ordinary as DNS queries. Going forward I will always check DNS logs first and look for repeated queries to the same domain at regular intervals before reviewing authentication events. Typosquatting is also subtle, a single character difference like microsofft versus microsoft is easy to miss when scanning logs quickly, which is why character-by-character domain verification is important.

Lessons Learned

Always read raw logs chronologically before running queries. The beaconing was visible on first read if I had been looking for it. The queries began at 00:03 while the brute force did not start until 00:14. That 11-minute gap should have prompted the question of what was happening before the attack started.

Typosquatting detection requires attention to detail. Attackers register domains that look almost identical to legitimate ones. Checking domains character by character against known legitimate domains is a habit that should become automatic.

Archive size correlation is a reliable exfiltration proof technique. When an archive is created and an outbound connection of the same size follows immediately, that is confirmed exfiltration no assumption needed.

Multiple persistence mechanisms mean a thorough cleanup is essential. Finding one backdoor does not mean the machine is clean. Always search for scheduled tasks, registry Run keys and new accounts created during the attack window.

Port 4444 means an active attacker. An outbound connection on port 4444 should trigger immediate escalation because it indicates the attacker has live remote control of the machine right now, not historical access.

Personal Reflection

This was the most challenging and rewarding investigation I have completed in this selfdirected programme. It exposed me to a pre-attack phase I had not encountered in the previous two cases, pushed me to think beyond reactive alert investigation and helped me understand that sophisticated attackers operate in stages with each phase enabling the next. The fact that I initially missed the beaconing was humbling but ultimately more valuable than getting it right immediately understanding what you miss and why is what makes you a better analyst over time.